is subjected to low light intensity, its photosynthesis is reduced. Thus, low light intensity is referred as stress and reduction of photosynthesis is referred as strain. Biological strains are of two types; Elastic biological strain and Plastic biological strain. If the reaction of plant function is temporary and when it returns to its original state it is called **elastic biological strain**. Example: Temporary wilting. If the reaction is permanent and the plant function does not return to the normal state it is called **plastic biological strain**. Example: Permanent wilting. Some plants get adapted to stress condition and are not adversely affected by stress. Such plants are called **stress resistant** or **stress tolerant** plants. Example: Mangroves. Some plants cannot face stress and they pass their adverse period in dormant state and so they are called **stress enduring plants**. Ephemeral plants are short lived desert plants, which complete their life cycle during the seasonal rains before the onset of dry season. These ephemeral plants are called **stress escapers**. Stress in plants can be classified as given in figure 15.26.

1. Biotic Stresses

These are adverse effects on plants caused by other living organisms such as viruses, bacteria, fungi, parasites, insects, weeds and competing plants. Biotic environmental stress is also caused due to the activity of man by cutting herbs and trees, twigs for fodders, fuels and agricultural purposes. The biotic stresses caused by bacteria, fungi and nematodes that are ever present in the environment are called potential biotic stresses. These are divided into two types. They are:

i) Allelopathy

An organism producing one or more biochemical substances that greatly influence the germination, growth and reproduction of other organisms is called **Allelopathy**. These biochemicals are known as **allelochemicals**. They are beneficial (positive allelopathic) or detrimental (negative allelopathic). These allelochemicals are obtained from leaf after leaching on the ground and also from roots. The term allelopathy is from Greek words *allelon*-each other and *pathos*-to suffer and first used in 1937 by **Hans Molisch**. Allelopathic effect may occur with weeds on crops and *vice versa* (Figure 15.27).

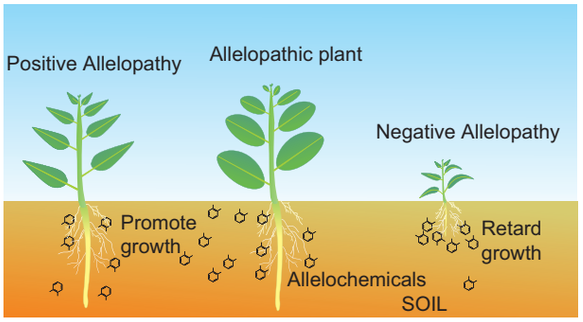


Figure 15.27: Allelopathy in plants

One of the most famous allelopathic plants is Black walnut (*Juglans nigrum*). The chemical which is present in Black walnut is Juglone and it is a respiratory inhibitor. Solanaceous plants such as tomato, capsicum and eggplant are susceptible to juglone. These plants when exposed to these allelochemicals exhibit symptoms such as wilting, chlorosis and death.

Check your grasp! Are all plants allelopathic? Can allelopathic chemicals affect animals and humans?

Tree of heaven (*Ailanthus altissima*) is a recent addition to the list of allelopathic trees. **Ailanthone** an allelochemical extracted from the root of *Ailanthus* acts as potent herbicide. In *Sorghum* plant the allelochemical sorgolone possess allelopathic activity. It is found in root exudates of most *Sorghum* species. Root exudation of maize inhibits the growth of some weeds such as *Chenopodium album* and *Amaranthus retroflexus*. The seed exudates of oat *(Avena fatua*) affect the germination of wheat seedling.

ii. Pathogenecity

The effect of microbes that cause diseases in plants. Example: *Xanthomonas citri*

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